

6(a)	volt / ampere	B1
6(b)(i)	1. $I = 1.8 + 0.60$ $= 2.4 \text{ A}$	A1
	2. $(8.0 \times 0.60) = 1.8 \times (2.0 + R_Z)$	C1
	$R_Z = 0.67 \Omega$	A1
	3. $E - (2.4 \times 1.5) = (0.60 \times 8.0)$ or $E - (2.4 \times 1.5) = 1.8 \times (2.0 + 0.67)$ or $E = 2.4 \times [1.5 + (8.0 \times 2.67) / (8.0 + 2.67)]$	C1
	$E = 8.4 \text{ V}$	A1
6(b)(ii)	1. $R = \rho L / A$ or $R \propto 1 / A$	C1
	ratio $= R_Y / R_X = 2.0 / 8.0$ $= 0.25$	A1
	2. $I \propto Av$ or $I_X / I_Y = A_X v_X / A_Y v_Y$	C1
	ratio $= (0.60 / 1.8) \times (1 / 0.25)$ $= 1.3$	A1